MidTerm Exam: AI in HC Glossary

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Introduction

This glossary covers twenty terms from the AI and healthcare literature, focusing on neonatal care applications. Each entry offers a definition, an example from clinical or research practice, and an explanation of why it matters. The selection includes both foundational concepts and specialized vocabulary because a glossary is only helpful when it links technical details to practical implications.

Body

AI in Healthcare Glossary

1. Algorithmic bias

Definition: Algorithmic bias is systematic error in a model's design, training data, or evaluation that causes uneven performance across patient groups, often reflecting disparities already encoded in the health records used for training (Hasanzadeh et al., 2025).

Example in healthcare: A NICU deterioration model developed at one academic center may underperform at a community hospital where demographics and documentation practices differ from the training environment (Sullivan et al., 2024).

Significance: Biased tools can underestimate risk for populations already receiving fewer resources, compounding the disparities healthcare systems aim to reduce (Hasanzadeh et al., 2025).

2. Artificial intelligence (AI)

Definition: Artificial intelligence encompasses computational methods that perform tasks ordinarily requiring human cognition, including pattern recognition, language interpretation, and decision-making under uncertainty (Keles & Bagci, 2023).

Example in healthcare: In neonatal care, AI applications range from analyzing chest radiographs for lung disease to extracting structured information from free-text notes (Sullivan et al., 2024).

Significance: In the NICU, where monitoring generates volumes exceeding what any clinician can process in real time, AI can surface patterns that might otherwise go unnoticed, though its value depends on responsible validation and integration (Keles & Bagci, 2023).

3. Area under the receiver operating characteristic curve (AUROC)

Definition: AUROC quantifies how well a model separates patients who experience an outcome from those who do not, ranging from 0.5 to 1.0 (Chou et al., 2024).

Example in healthcare: A deep learning model predicting bronchopulmonary dysplasia from neonatal chest radiographs reported an AUROC of 0.94 (Chou et al., 2024).

Significance: A high AUROC demonstrates separation but does not confirm clinical readiness, because real-world performance also depends on calibration and external validation (Schwabe et al., 2024).

4. Calibration

Definition: Calibration describes whether a model's predicted probabilities match the actual frequency of outcomes. A well-calibrated model estimating 20% sepsis risk should see that outcome in roughly one of every five similar cases (Schwabe et al., 2024).

Example in healthcare: Calibration analysis of a NICU sepsis model compares predicted probabilities against confirmed infection rates within each risk tier (Schwabe et al., 2024).

Significance: Clinicians act on the probability itself, so a model reporting a 40% risk when the true likelihood is 10% leads to unnecessary interventions. In NICU care, the reliability of probability is as consequential as discrimination (Schwabe et al., 2024).

5. Clinical decision support (CDS)

Definition: Clinical decision support is technology that synthesizes patient-specific data and delivers evidence-based recommendations or risk assessments at the point of care (Sullivan et al., 2024).

Example in healthcare: A NICU CDS tool might integrate vital-sign trends with laboratory results to generate a deterioration score, flagging neonates before overt instability emerges (Keles & Bagci, 2023).

Significance: Well-designed CDS standardizes responses and reduces cognitive burden, though the challenge is avoiding excessive alerts that desensitize clinicians to genuine warnings (Sullivan et al., 2024).

6. Computer vision

Definition: Computer vision enables machines to interpret and extract information from visual data such as medical images and video (Keles & Bagci, 2023).

Example in healthcare: Algorithms have been applied to neonatal chest radiographs to identify pneumothorax, atelectasis, and bronchopulmonary dysplasia (Chou et al., 2024).

Significance: Automated analysis can extend subspecialty expertise to facilities without on-site radiologists, enabling community NICUs to access interpretation quality comparable to tertiary centers (Keles & Bagci, 2023).

7. Convolutional neural network (CNN)

Definition: A convolutional neural network is a deep learning architecture that processes images through learnable filters detecting spatial features across successive layers (Keles & Bagci, 2023).

Example in healthcare: CNNs have analyzed neonatal chest radiographs to predict bronchopulmonary dysplasia, capturing visual features difficult to quantify manually (Chou et al., 2024).

Significance: CNNs learn directly from pixel data without manual feature extraction, making them well suited to NICU imaging where specialist access remains a bottleneck (Keles & Bagci, 2023).

8. Data drift

Definition: Data drift occurs when input data after deployment differ from training data, often due to changes in clinical practice, demographics, or equipment (Wong & Sussman, 2025).

Example in healthcare: A NICU model trained on one monitoring platform may experience drift when hardware is upgraded, because signal characteristics change even when clinical meaning does not (Wong & Sussman, 2025).

Significance: Drift degrades accuracy without an obvious error, underscoring the need for ongoing monitoring rather than relying solely on the initial validation study (Wong & Sussman, 2025).

9. Deep learning

Definition: Deep learning employs neural networks with multiple layers to learn complex, hierarchical representations from data (Keles & Bagci, 2023).

Example in healthcare: Deep learning has been applied to neonatal images and physiologic time-series to predict mortality, sepsis, and chronic lung disease (Keles & Bagci, 2023; Chou et al., 2024).

Significance: Its performance on imaging often exceeds simpler models, though reduced transparency and the difficulty of explaining conclusions remain barriers to clinical trust (Sullivan et al., 2024).

10. Digital biomarker

Definition: A digital biomarker is a physiologic or behavioral measure captured through a digital device, such as a wearable, that assesses health status or estimates risk (Jasinski et al., 2024).

Example in healthcare: Maternal heart rate variability from a wearable has been studied as a biomarker for preterm birth, aiming to identify risk before symptoms emerge (Jasinski et al., 2024).

Significance: Digital biomarkers extend monitoring beyond scheduled visits, potentially enabling earlier detection of complications, though validation across populations and devices remains essential (Jasinski et al., 2024).

11. Electronic health record (EHR)

Definition: An electronic health record is a comprehensive digital archive of clinical information, including diagnoses, medications, lab results, imaging, and narrative documentation (Sullivan et al., 2024).

Example in healthcare: Neonatal AI models rely on EHR data such as ventilator settings, medication exposures, and gestational age to assemble training datasets (Keles & Bagci, 2023).

Significance: EHR quality directly determines model reliability, and missingness or inconsistent coding weakens performance, making data governance a prerequisite (Schwabe et al., 2024; Duda et al., 2022).

12. Explainable artificial intelligence (XAI)

Definition: Explainable AI encompasses methods that make a model's reasoning understandable to clinicians who must decide whether to act on a prediction (Hasanzadeh et al., 2025).

Example in healthcare: Saliency maps on neonatal chest radiographs show which lung regions influenced a prediction, letting clinicians assess whether the model's focus aligns with known pathology (Chou et al., 2024).

Significance: When clinicians cannot understand why a tool generated a score, it becomes either blindly followed or ignored. In the NICU, explainability is a condition for meaningful adoption (Hasanzadeh et al., 2025).

13. **Fast Healthcare Interoperability Resources (FHIR)**

Definition: FHIR is a data standard from Health Level Seven International that defines how healthcare information is structured and exchanged between electronic systems (Duda et al., 2022).

Example in healthcare: FHIR-based tools can extract neonatal laboratory values and medication orders from an EHR and structure them for AI development without manual chart abstraction (Duda et al., 2022).

Significance: Without interoperability standards, data from different hospitals arrive in incompatible formats, limiting the reach of otherwise promising AI tools beyond a single institution (Duda et al., 2022).

14. Foundation model

Definition: A foundation model is trained on broad data in a self-supervised manner and can be adapted to multiple tasks without entirely new training for each (Huang et al., 2025).

Example in healthcare: NeoCLIP was trained on over 20,000 neonatal radiographs and can detect fifteen pathologies and five medical devices relevant to NICU care (Huang et al., 2025).

Significance: A single model with generalizable representations can be fine-tuned for different conditions, which matters in neonatology where labeled datasets per condition tend to be small (Huang et al., 2025).

15. Generative artificial intelligence

Definition: Generative AI refers to systems that produce new content such as text or summaries by learning patterns from training data (Sullivan et al., 2024).

Example in healthcare: Generative AI has been explored for drafting documentation, summarizing NICU notes, and creating plain-language discharge instructions (Sullivan et al., 2024).

Significance: Efficiency benefits are tangible where clinicians spend disproportionate time documenting, though outputs require verification because generative models can produce plausible text containing factual errors (Sullivan et al., 2024).

16. Risk stratification

Definition: Risk stratification categorizes patients by their estimated likelihood of adverse outcomes, enabling targeted allocation of monitoring and interventions (Sullivan et al., 2024).

Example in healthcare: AI can classify preterm infants into respiratory risk tiers using gestational age, birth weight, and physiologic trends, guiding surveillance decisions (Sullivan et al., 2024).

Significance: Distributing attention equally may over-monitor stable neonates while under-monitoring those deteriorating. When stratification relies on biased data, it can disadvantage the populations most in need (Hasanzadeh et al., 2025).

Conclusion

In conclusion, compiling this glossary clarified that AI in healthcare depends on more than algorithmic sophistication. Data quality, validation, interoperability, and transparency all condition whether a tool that performs well in research will function responsibly at the bedside. The NICU illustrates this clearly because its continuous monitoring, high-acuity decisions, and vulnerable patients amplify both the promise and the risk of every AI application.

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